
Algorithm 1 Minibatch stochastic gradient descent training of generative adversarial nets. The number of steps to apply to the discriminator, k , is a hyperparameter. We used $k = 1$, the least expensive option, in our experiments.

for number of training iterations **do**

for k steps **do**

- Sample minibatch of m noise samples $\{z^{(1)}, \dots, z^{(m)}\}$ from noise prior $p_g(z)$.
- Sample minibatch of m examples $\{x^{(1)}, \dots, x^{(m)}\}$ from data generating distribution $p_{\text{data}}(x)$.
- Update the discriminator by ascending its stochastic gradient:

$$\nabla_{\theta_d} \frac{1}{m} \sum_{i=1}^m \left[\log D(x^{(i)}) + \log \left(1 - D(G(z^{(i)})) \right) \right].$$

end for

- Sample minibatch of m noise samples $\{z^{(1)}, \dots, z^{(m)}\}$ from noise prior $p_g(z)$.
- Update the generator by descending its stochastic gradient:

$$\nabla_{\theta_g} \frac{1}{m} \sum_{i=1}^m \log \left(1 - D(G(z^{(i)})) \right).$$

end for

The gradient-based updates can use any standard gradient-based learning rule. We used momentum in our experiments.

4.1 $p_g = p_{\text{data}}$ 的全局最优性

我们首先考虑对于任意给定生成器 G 的最优判别器 D

命题 1. 对于固定的 G ，最优判别器 D 为

$$D_G^*(x) = \frac{p_{\text{data}}(x)}{p_{\text{data}}(x) + p_g(x)} \quad (2)$$

$V(G, D)$ 证明。给定任意生成器 G ，判别器 D 的训练准则是最大化以下量

$$\begin{aligned} V(G, D) &= \int_{\mathbf{x}} p_{\text{data}}(\mathbf{x}) \log(D(\mathbf{x})) d\mathbf{x} + \int_{\mathbf{z}} p_g(\mathbf{z}) \log(1 - D(G(\mathbf{z}))) d\mathbf{z} \\ &= \int_{\mathbf{x}} p_{\text{data}}(\mathbf{x}) \log(D(\mathbf{x})) + p_g(\mathbf{x}) \log(1 - D(\mathbf{x})) d\mathbf{x} \end{aligned} \quad (3)$$

对于任意 $(a, b) \in \mathbb{R}^2 \setminus \{0, 0\}$ ，函数 $y \rightarrow a \log(y) + b \log(1 - y)$ 在 $[0, 1]$ 区间内于 $\frac{a}{a+b}$ 处取得最大值。判别器无需在 $\text{Supp}(p_{\text{data}}) \cup \text{Supp}(p_g)$ 之外定义，证毕。 $\square$

注意， D 的训练目标可以解释为最大化估计条件概率 $P(Y = y | x)$ 的对数似然，其中 Y 表示 x 是来自 p_{data} （当 $y = 1$ ）还是来自 p_g （当 $y = 0$ ）。式 1 中的极小极大博弈现在可以重新表述为：

$$\begin{aligned} C(G) &= \max_D V(G, D) \\ &= \mathbb{E}_{\mathbf{x} \sim p_{\text{data}}} [\log D_G^*(\mathbf{x})] + \mathbb{E}_{\mathbf{z} \sim p_g} [\log(1 - D_G^*(G(\mathbf{z})))] \\ &= \mathbb{E}_{\mathbf{x} \sim p_{\text{data}}} [\log D_G^*(\mathbf{x})] + \mathbb{E}_{\mathbf{x} \sim p_g} [\log(1 - D_G^*(\mathbf{x}))] \\ &= \mathbb{E}_{\mathbf{x} \sim p_{\text{data}}} \left[\log \frac{p_{\text{data}}(\mathbf{x})}{p_{\text{data}}(\mathbf{x}) + p_g(\mathbf{x})} \right] + \mathbb{E}_{\mathbf{x} \sim p_g} \left[\log \frac{p_g(\mathbf{x})}{p_{\text{data}}(\mathbf{x}) + p_g(\mathbf{x})} \right] \end{aligned} \quad (4)$$